



POSITIVE & NEGATIVE **NUMBERS**

NEGATIVE

POSITIVE



All numbers to the LEFT of zero
are NEGATIVE NUMBERS.

All numbers to the RIGHT of
zero are POSITIVE NUMBERS.

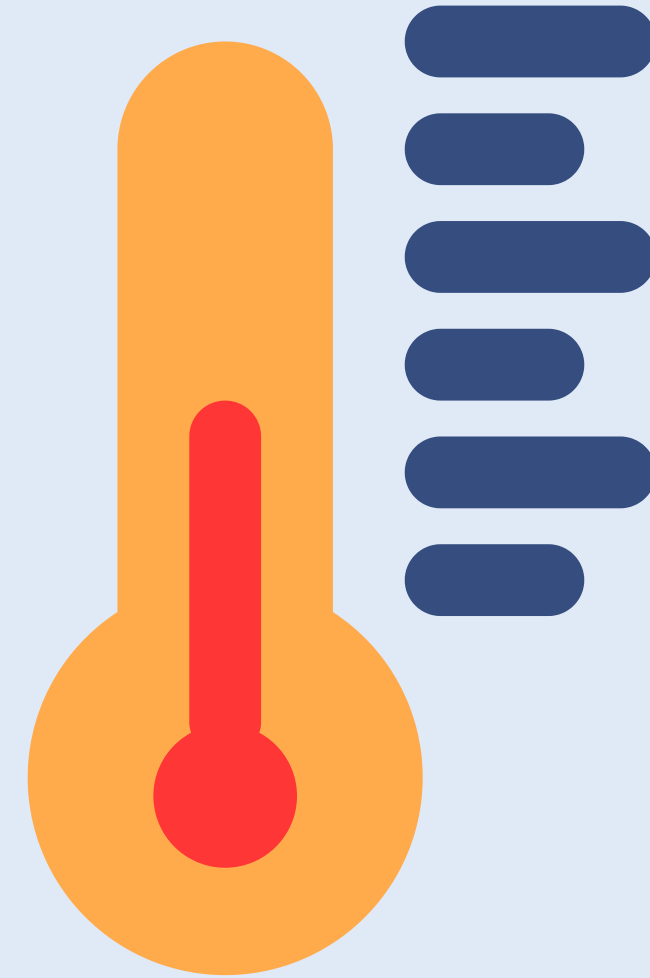


**LET'S LOOK
AT SOME
EXAMPLES...**

TEMPERATURE

Temperatures above
zero are POSITIVE.

Temperatures below
zero are NEGATIVE.



WARMER

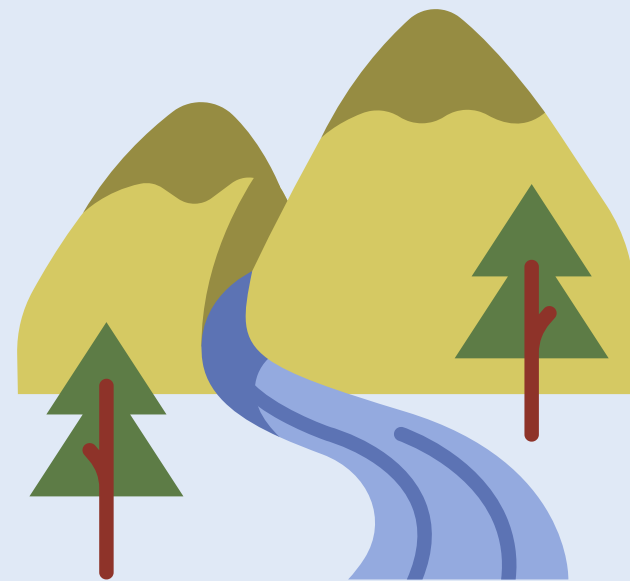
+20°

0°

-12°

COOLER

ELEVATION



ABOVE
SEA LEVEL

+800 ft

0 ft

-200 ft

BELOW
SEA LEVEL

Elevations above zero
are POSITIVE.

Elevations below zero
are NEGATIVE.

FINANCES

Values above zero are
POSITIVE ie. when money goes
into your bank account.

Values below zero are
NEGATIVE ie. when money
leaves your bank account.



CREDIT
(MONEY IN)

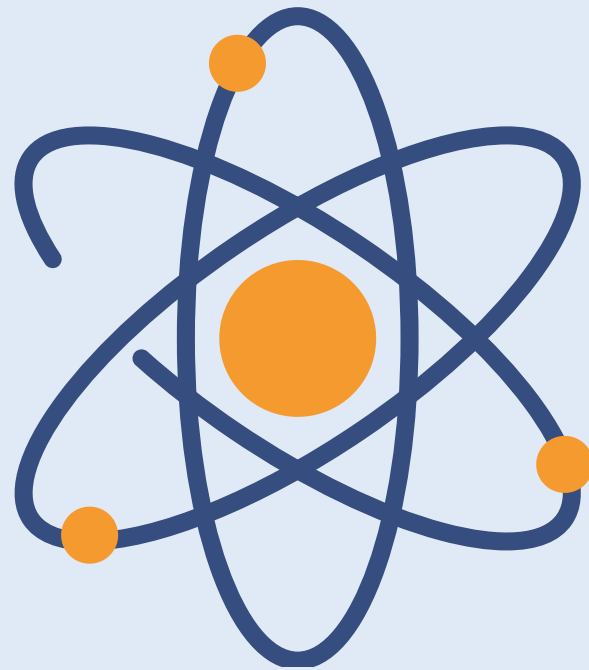
+\$50

\$0

-\$20

DEBIT
(MONEY OUT)

ELECTRIC CHARGE



POSITIVE
CHARGE

+2

0

-2

NEGATIVE
CHARGE

Electric charges above zero are POSITIVE. These are positive ions.

Electric charges below zero are NEGATIVE. These are negative ions.



WHAT ARE INTEGERS?

Integers and the
Number Line

DEFINITION OF INTEGERS

Integers consist of all positive whole numbers, negative whole numbers and zero.

-3, -5, 0, 1, 7, and 18 are integers.

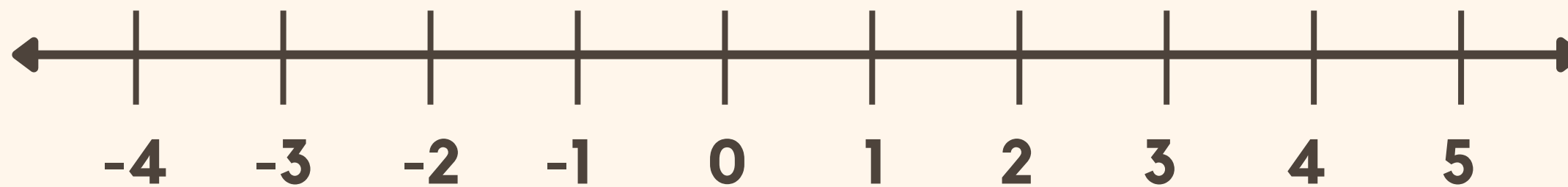
1.6, $\frac{1}{2}$, 5%, and -2.43 are not integers.

INTEGERS ON A NUMBER LINE

We can illustrate integers using a number line.

Fun Fact!
In a number line,
numbers are
arranged from
left to right in
increasing order.

Zero is neither
positive nor negative.



NEGATIVE INTEGERS

Negative numbers
are on the left of 0.

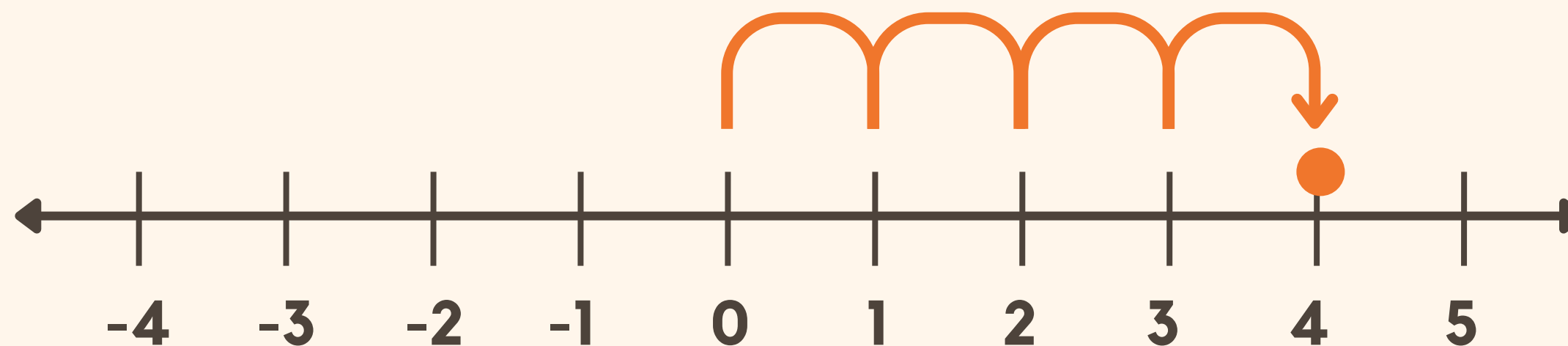
POSITIVE INTEGERS

Positive numbers
are on the right of 0.

PLOTTING INTEGERS ON A NUMBER LINE

Locate the corresponding point to the left or to the right of 0.

Example: Plot 4 on the number line.

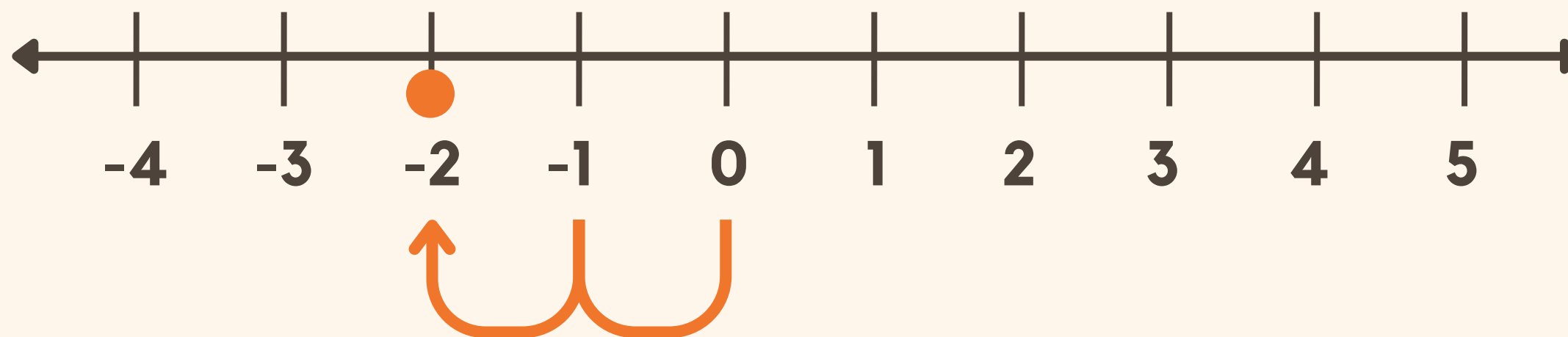


Because 4 is positive, we plot 4 units to the right of 0.

PLOTTING INTEGERS ON A NUMBER LINE

Locate the corresponding point to the left or to the right of 0.

Example: Plot -2 on the number line.



Because -2 is negative, we plot 2 units to the left of 0.

LET'S TRY

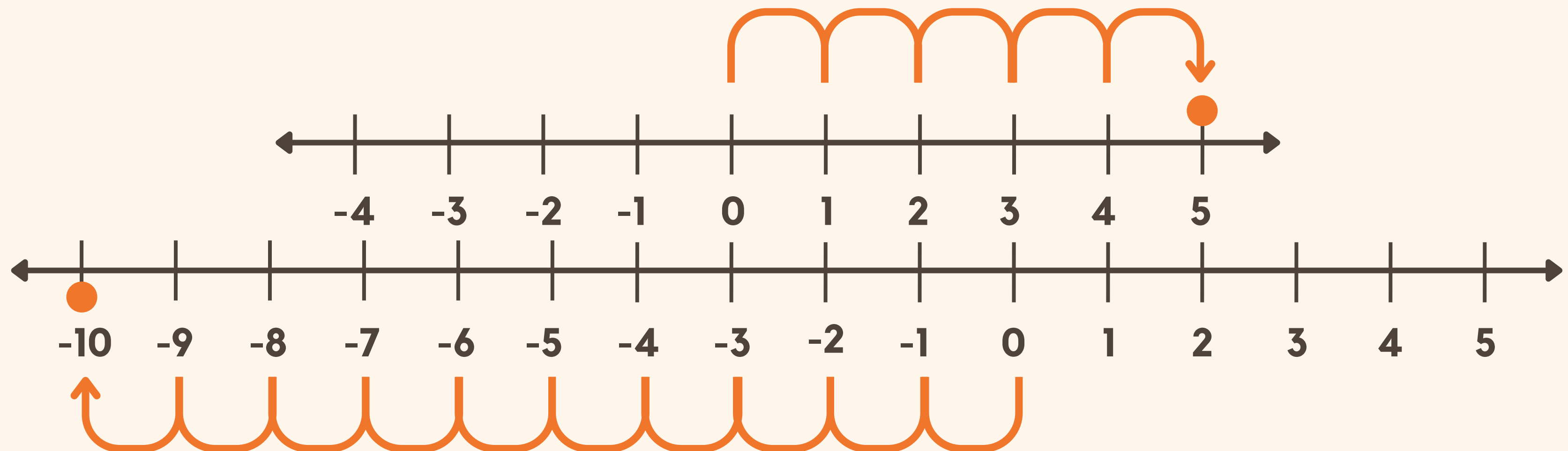
Find a pair and recall the previous question.

From 0 degrees Celsius, the temperature increased by 5 degrees in the morning. In the evening, it decreased by 10 degrees. How can we represent these values using integers?



LET'S REPRESENT

Increase of temperature by 5 degrees



Decrease of temperature by 10 degrees

ASCENDING ORDER

To write numbers in ascending order, we arrange them from smallest to biggest.

50,000 is the smallest number. It is in the first position.

1,850,000 is the biggest number. It is in the last position.

We write the number in the increasing order as follows:

50,000

75,000

150,200

1,850,000



DESCENDING ORDER

To write numbers in descending order, we arrange them from biggest to smallest.

1,850 is the biggest number. It is in the first position.

50,000 is the smallest number. It is in the last position.

We write the numbers in the decreasing order as follows:

1,850,000

150,200

75,000

50,000



Rational and Irrational Numbers

Rational numbers

A number is rational if it can be expressed as a fraction. It includes positive and negative numbers including 0.

7 -6 $\frac{1}{9}$

Irrational numbers

Irrational numbers cannot be expressed as a simple fraction. When written as a decimal, they are non recurring and do not terminate.

π 2.6173

**Great
Work
Today!**

